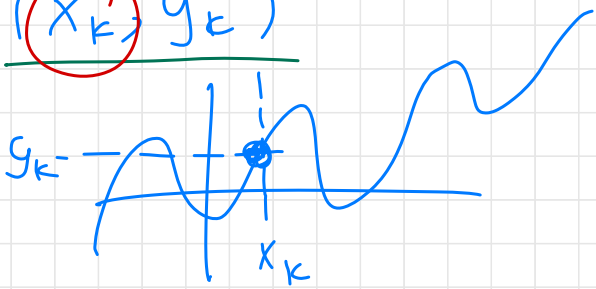
Goal: convert $A(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1}$

to n samples (x_k, y_k)

$y_k = A(x_k)$

x_k distinct



why? $C(x) = A(x) \cdot B(x)$ in $O(n \log n)$ time

$O(n^2)$: plug in n x_k vals $x_0, x_1, \ldots, x_{n-1}$

$$\boxed{\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & \cdots \; n-1 \end{array} \\ \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & & & & \\ 1 & x_{n-1} & x_{n-1}^2 & \cdots & x_{n-1}^{n-1} \end{pmatrix} \end{array}} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{n-1} \end{pmatrix} \leftarrow$$

$$V \cdot A = Y$$

Divide + Conquer

want $A(x)$ for $x \in X$

① Divide into even and odd coeffs

$A_{even}(x) = a_0 + a_2 x + a_4 x^2 + \cdots = \sum_{k=0}^{\sim n/2} a_{2k} x^k = \langle a_0, a_2, a_4, \ldots \rangle$

$A_{odd}(x) = a_1 + \boxed{a_3 x} + a_5 x^2 + \cdots = \sum_{k=0}^{n/2} a_{2k+1} x^k = \langle a_1, a_3, a_5, \ldots \rangle$

② Conquer

compute $\boxed{A_{even}(y)}$, $\boxed{A_{odd}(y)}$ for $y \in \boxed{X^2}$

$= \{x^2 : x \in X\}$

③ Combine

$A(x) = \boxed{A_{even}(x^2)} + x \boxed{A_{odd}(x^2)}$ $\boxed{\text{for } x \in X}$

$\cancel{(a_2 + a_3)x}$

$a_2 x^2 + a_3 x^3 \leftarrow$

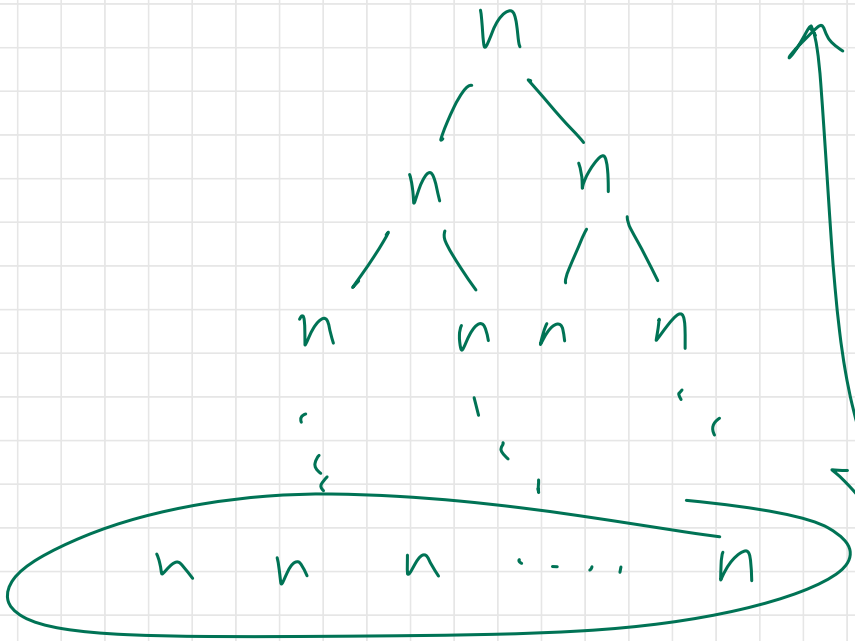
$A_{odd}(x) = a_1 x + a_3 x^2 + a_5 x^3 + \cdots$

$a_1 x + a_3 x^3 + a_5 x^5 + \cdots$

④ Runtime

$T(\underset{\#\text{ coeffs}}{n}, \underset{\#\text{ samples}}{|X|}) = \underbrace{2T(n/2, |X|)}_{\text{recursive calls}} + \underbrace{\overset{①}{\boxed{n}} + \overset{①, ③}{\boxed{|X|}}}_{\text{non-rec. work}}$

$\uparrow$ n



$\log n$

$2^{\log_2 n} = n$

$O(n^2)$

$|X^2| = |X|/2$

$T(n) = 2T(n/2) + n \Rightarrow T(n) = O(n \log n)$

$|X| = 1 \quad X = \{1\}$

square

$|X| = 2 : X = \{-1, 1\}$ so $X^2 = \{(-1)^2, (1)^2\}$
$= \{1, 1\} = \{1\}$

idea 1:

$|X| = 4 : X = \{-2, -1, 1, -2\}$ so
$X^2 = \{1, 4\}$

idea 2:

$|X| = 4$: $X = \boxed{\{1, -1, i, -i\}}$

$i = \sqrt{-1}$

$1^2 = 1$

$(-1)^2 = 1$

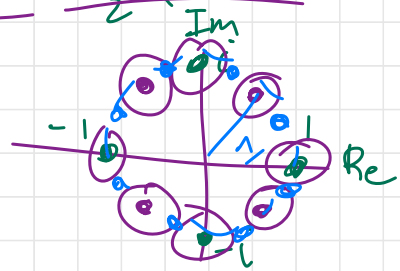
$(i^2) = (\sqrt{-1})^2 = -1$

$-(i)^2 = (\sqrt{-1})^2 = -(-1) = 1$

$8\sqrt{1}$

8th roots of unity

$|X| = 8$: $X = \{\pm\frac{\sqrt{2}}{2}(i+1), \pm\frac{\sqrt{2}}{2}(1-i), \pm i, \pm 1\}$



n^{th} roots of unity are collapsing

$|X^2| = |X|/2$ or

$|X| = 1$

Discrete Fourier Transform (DFT)

$DFT(A(x)) = A(x)$ at n^{th} roots of unity

$= V \cdot A$ for $X_k \in$ set of nth roots of unity

Fast Fourier Transform (FFT)

$O(n \log n)$ alg to compute $DFT(A(x))$

Fast Polynomial Mult (A, B):

$A^* = \underline{FFT(A)}$

$B^* = \underline{FFT(B)}$

$\underline{C^*_k} = A^*_k \cdot B^*_k \quad \forall k$ ← sample values for C

$C = IFFT(C^*)$

$IFFT(P^*)$

$P' = \dfrac{FFT(P^*)}{n}$